

System Under Test (SUT)

Heterogeneous Embedded SoC / Dual-Brain Topology

1. Sensor Frame Ingest (GPIO 1):

Toggled inside camera driver ISR on DMA start.

2. MPU Policy Inference Emit (GPIO 2):

Toggled in kernel space when writing inter-core mailbox.

3. MCU Enforcement Veto / Pass (GPIO 3):

Toggled in 1 kHz timer ISR upon CBF solver exit.

4. Gate Driver Inverter Output (PWM Pin):

Direct hardware timer register asserting motor phase.

Probe 1: Ingest

Probe 2: Inference

Probe 3: Enforce

Probe 4: Coil Current

External Ground-Truth Metrology

Multi-Channel Logic Analyzer & Oscilloscope Current Shunt

CH 1 (Digital): Transduction Pulse → True $t_{\text{transduce}}$

CH 2 (Digital): Proposal Generated → True t_{infer}

CH 3 (Digital): MCU Veto / Pass → True t_{enforce}

CH 4 (Analog): Shunt Resistor → Stator Coil Current Rise

Hardware Invariant: Zero Observer Jitter (< 10 ns error).